

Optimal Stopping Limits in Permutation Streams

Analytical Derivation and Asymptotic Limits of the Secretary Problem

A Rigorous Analytical Derivation, Stochastic Analysis, and Empirical Verification

Vedant Raut

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Abstract

This paper presents a formal analytical derivation and stochastic evaluation of optimal stopping limits in finite and continuous decision processes under discrete uniform permutation streams.

By modeling sequential candidate selection under incomplete information across N discrete arrivals, we establish the discrete probability sum $P(r) = \frac{r-1}{N} \sum_{i=r}^N \frac{1}{i-1}$ for a sample cutoff index r .

We evaluate its continuous asymptotic limit $P(x) = -x \ln(x)$ as applicant pool size $N \rightarrow \infty$ and prove via calculus that the global maximum success rate strictly converges to $x^* = 1/e \approx 36.79\%$.

Furthermore, we discuss applications in algorithmic trade execution and evaluate a companion interactive web simulation.

My Live Simulation Engine : [Access Interactive Simulation Web Application](#)

Core Math Domain: Probability & Dynamic Optimization

Primary Focus: Information Thresholds in Discrete Order Statistics

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1 Problem Formulation & Intuitive Model

Imagine interviewing N distinct candidates sequentially for a single position. The applicants arrive in a uniformly random order, and you can rank candidates relative to those interviewed previously.

The decision framework is governed by four strict operational rules:

- i. Candidates are evaluated one by one in a strictly sequential stream.
- ii. Upon interviewing candidate i , you must make an immediate, irrevocable decision to accept or reject them.
- iii. Rejected candidates cannot be recalled under any circumstances.
- iv. The process terminates instantly upon selecting a candidate, with the sole objective of maximizing the probability of hiring the single best overall candidate out of N .

1.1 Primary Investigation Objectives

The core analytical goals of this research paper are defined across two operational domains:

- **Analytical Derivation:** Derive the exact cutoff strategy r^* that balances sample gathering with selection, proving that $P(x) = -x \ln(x)$ peaks at $1/e$.
- **Stochastic Verification:** Simulate Monte Carlo permutation streams to verify that empirical success rates match the theoretical 36.79% asymptotic bound.

1.2 The Two-Phase Threshold Strategy

To solve this optimization dilemma, we split the search process into two distinct operational phases:

Phase 1 (Observation): Automatically reject the first $r - 1$ candidates to establish a baseline quality benchmark $B = \max_{1 \leq j < r} Q_j$.

Phase 2 (Decision): Hire the very first candidate $i \geq r$ whose quality strictly exceeds the Phase 1 benchmark ($Q_i > B$).

2 Theoretical Proof & Mathematical Derivation

To determine the optimal cutoff position r , we calculate the probability $P(r)$ that the absolute best candidate in the entire pool is successfully selected.

Suppose the overall best candidate is positioned at index i (where $i \geq r$). This candidate is selected if and only if the best candidate among the first $i - 1$ applicants landed within the initial observation window $[1, r - 1]$.

$$P(\text{Candidate } i \text{ is selected}) = \frac{r - 1}{i - 1} \quad (1)$$

Equation (1) establishes the probability that candidate i is chosen given that they are the absolute best applicant.

Summing across all possible candidate positions i from r to N yields the discrete overall success probability $P(r)$.

$$P(r) = \sum_{i=r}^N P(\text{Candidate } i \text{ is best}) \cdot P(\text{Candidate } i \text{ is selected}) \quad (2)$$

Equation (2) applies the Law of Total Probability over all possible positions i .

Substituting the uniform position probability $1/N$ and equation (1) into this sum isolates constant factors.

$$P(r) = \sum_{i=r}^N \frac{1}{N} \cdot \left(\frac{r - 1}{i - 1} \right) = \frac{r - 1}{N} \sum_{i=r}^N \frac{1}{i - 1} \quad (3)$$

Equation (3) expresses the exact discrete probability formula $P(r)$ for any finite pool size N .

2.1 Continuous Asymptotic Limit as $N \rightarrow \infty$

We analyze the limiting behavior as applicant size $N \rightarrow \infty$. Define the continuous cutoff fraction $x = \lim_{N \rightarrow \infty} \frac{r}{N}$ and let $t = \frac{i}{N}$ with step differential $dt = \frac{1}{N}$.

The discrete summation converts directly into a continuous Riemann integral over interval $[x, 1]$.

$$P(x) = \lim_{N \rightarrow \infty} \frac{r - 1}{N} \sum_{i=r}^N \frac{1}{(i - 1)/N} \cdot \frac{1}{N} = x \int_x^1 \frac{1}{t} dt \quad (4)$$

Equation (4) formulates the continuous probability function $P(x)$.

We evaluate the definite integral using the fundamental theorem of calculus.

$$\int_x^1 \frac{1}{t} dt = [\ln(t)]_x^1 = \ln(1) - \ln(x) = -\ln(x) \quad (5)$$

Equation (5) calculates the integral antiderivative $\ln(t)$, evaluating to $-\ln(x)$.

Substituting this result back into equation (4) defines the continuous probability curve $P(x)$.

$$P(x) = -x \ln(x) \quad (6)$$

Equation (6) establishes the continuous success probability density curve over $x \in (0, 1]$.

2.2 Optimization via First Principles Calculus

To find the optimal threshold fraction x^* , we differentiate $P(x)$ with respect to x using the product rule.

$$P'(x) = \frac{d}{dx}[-x \ln(x)] = -1 \cdot \ln(x) - x \cdot \left(\frac{1}{x}\right) = -\ln(x) - 1 \quad (7)$$

Equation (7) evaluates the first derivative. Setting $P'(x) = 0$ locates the critical point x^* .

$$-\ln(x^*) - 1 = 0 \implies \ln(x^*) = -1 \implies x^* = e^{-1} = \frac{1}{e} \quad (8)$$

Equation (8) solves for the optimal threshold: $x^* = 1/e \approx 0.367879$ (or 36.79%).

Substituting $x^* = 1/e$ back into equation (6) yields the maximum theoretical success probability.

$$P(x^*) = -\left(\frac{1}{e}\right) \ln\left(\frac{1}{e}\right) = -\frac{1}{e}(-1) = \frac{1}{e} \approx 36.79\% \quad (9)$$

Equation (9) completes the proof: both the optimal observation sample fraction x^* and the maximum achievable success probability $P(x^*)$ equal strictly $1/e$.

3 Computational Simulation & Asymptotic Error Analysis

In an empirical Monte Carlo simulation with M total trial permutations of N candidate ranks, we evaluate sample success counts across cutoff indices $r \in [1, N]$.

3.1 Convergence of Empirical Peak Rates

- **Under-sampling** ($r < \lfloor N/e \rfloor$): Terminating observation too early leads to premature selection of sub-optimal candidates, lowering empirical success below $1/e$.
- **Optimal Threshold** ($r^* = \lfloor N/e \rfloor$): Sampling approximately 36.8% of candidates achieves the maximum theoretical success rate $P(r^*) \approx 36.79\%$.
- **Over-sampling** ($r > \lfloor N/e \rfloor$): Prolonging observation increases the risk that the absolute best candidate appears in Phase 1 and is passed over, forcing selection of late mediocre candidates.

$$\lim_{N \rightarrow \infty} P(\text{Select Best Candidate} \mid r = \lfloor N/e \rfloor) = \frac{1}{e} \quad (10)$$

Equation (10) confirms asymptotic convergence to $1/e$ as sample size $N \rightarrow \infty$.

- **Theoretical Target:** Closed-form threshold fraction $x^* = 1/e \approx 0.367879$ and peak success probability $P(x^*) = 36.79\%$.
- **Empirical Convergence:** Simulations across $M = 100,000$ random permutations confirm that peak success aligns with $r^* = \lfloor N/e \rfloor$.

4 Applications and Practical Observations

Optimal stopping theory in discrete order statistics extends far beyond hiring models, serving as a core foundation for quantitative execution engines:

4.1 Optimal Order Execution & Market Impact

Algorithmic execution algorithms (such as TWAP/VWAP execution blocks) apply optimal stopping thresholds to route trade orders across dynamic order books, balancing price impact against execution risk over a fixed trading window.

4.2 American Option Exercise Boundaries

Valuing American-style derivatives requires identifying optimal stopping boundaries where the immediate payoff from early exercise strictly exceeds the continuation value of holding the contract.

4.3 High-Frequency Bid-Ask Matching Engines

Automated market-making engines utilize continuous point process stopping models to adjust liquidity quotes and optimize inventory management under high-frequency order flow uncertainty.

5 Interactive Web Simulation Evaluation

To complement this theoretical monograph, an interactive single-file web application was developed: `secretary_problem_sim.html`.

- **Live Interactive Application:** <https://optimalstoppingsecretaryproblem.vercel.app/>
- **GitHub Source Code Repository:** <https://github.com/wheyyydant/math-drafts-and-si>
- **Real-Time Candidate Stream Visualizer:** Displays N sequential arrivals, highlighting Phase 1 observation bounds, rejected baseline candidates, and selected candidates.
- **Empirical Probability Curve Plot:** Renders the continuous $P(x) = -x \ln(x)$ curve dynamically, marking the theoretical $1/e$ cutoff line alongside real-time Monte Carlo convergence traces.
- **Academic Design Visuals:** Utilizes a high-contrast slate layout focused on candidate data density, avoiding generic dark template clutter.

6 Concluding Remarks

We have presented a complete mathematical derivation of optimal stopping limits in discrete uniform permutation streams, proving that the optimal observation cutoff fraction and maximum selection probability both converge to $1/e \approx 36.79\%$.

The continuous density function $P(x) = -x \ln(x)$ illustrates the trade-off between baseline sampling and timely decision-making in stochastic systems.

7 Mathematical Glossary

Optimal Stopping Theory:	A field of probability concerned with choosing the optimal time to take an action to maximize expected reward under uncertainty.
Order Statistics:	Sample values organized in ascending or descending magnitude, defining relative rankings across discrete random variables.
Sample Benchmark (B):	The maximum relative rank observed during the initial Phase 1 sample window $[1, r - 1]$.
Asymptotic Bound ($1/e$):	The limiting value $e^{-1} \approx 0.367879$ governing both optimal sample size and maximum success probability as $N \rightarrow \infty$.
Law of Total Probability:	A fundamental rule expressing the total probability of an outcome as the sum of conditional probabilities across distinct events.